

$$\begin{aligned}
y &= x \cdot W + x \cdot \Delta W \\
\Delta W &= A \cdot B \\
A &= \text{masked_A} \cdot A_t \\
B &= \text{masked_B} \cdot B_t
\end{aligned}$$

Neuron 3, 5 are frozen
LoRA rank equals to 2
 $[\Delta W]_{i,3} = 0, \forall i = 1, 2, 3, 4$
 $[\Delta W]_{i,5} = 0, \forall i = 1, 2, 3, 4$

$$\Delta W = \begin{bmatrix} \Delta W_{11} & \Delta W_{12} & \Delta W_{13} & \Delta W_{14} & \Delta W_{15} & \Delta W_{16} \\ \Delta W_{21} & \Delta W_{22} & \Delta W_{23} & \Delta W_{24} & \Delta W_{25} & \Delta W_{26} \\ \Delta W_{31} & \Delta W_{32} & \Delta W_{33} & \Delta W_{34} & \Delta W_{35} & \Delta W_{36} \\ \Delta W_{41} & \Delta W_{42} & \Delta W_{43} & \Delta W_{44} & \Delta W_{45} & \Delta W_{46} \end{bmatrix}$$

$$= \begin{bmatrix} a_{11} & a_{12} \\ a_{21} & a_{22} \\ a_{31} & a_{32} \\ a_{41} & a_{42} \end{bmatrix} \begin{bmatrix} b_{11} & b_{12} & b_{13} & b_{14} & b_{15} & b_{16} \\ b_{21} & b_{22} & b_{23} & b_{24} & b_{25} & b_{26} \end{bmatrix}$$

$$= \begin{bmatrix} \dots & a_{11} \cdot b_{13} + a_{12} \cdot b_{23} & \dots & a_{11} \cdot b_{15} + a_{12} \cdot b_{25} & \dots \\ \dots & a_{21} \cdot b_{13} + a_{22} \cdot b_{23} & \dots & a_{21} \cdot b_{15} + a_{22} \cdot b_{25} & \dots \\ \vdots & \vdots & \vdots & \vdots & \vdots \\ \dots & a_{41} \cdot b_{13} + a_{42} \cdot b_{23} & \dots & a_{41} \cdot b_{15} + a_{42} \cdot b_{25} & \dots \end{bmatrix}$$

